

GEN AI MINI PROJECT

AI-Powered Personal Study Assistant

A Generative AI Application for Summarisation, Concept Explanation, and Personalised Revision

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1. Topic

This project is about building an AI-Powered Personal Study Assistant, a Generative AI application that helps a student study more effectively. The idea is simple: instead of spending hours converting lecture notes and textbook material into something usable, the assistant does the heavy lifting – it summarises the content, explains the tricky parts in plain language, generates practice questions, prepares revision notes, and gives the student some direction on what to focus on next.

2. Problem Statement

Most students end up spending a disproportionate amount of time just organising their study material before they can even begin revising – turning long chapters into short notes, picking out the important definitions, and creating questions to test themselves. This process is repetitive and often inefficient, especially closer to exams when time is limited. A GenAI-based assistant can take over this preparatory work, converting raw study material into structured, easy-to-revise resources within minutes rather than hours.

3. Objectives

The assistant is designed around a small set of practical goals:

- Summarise lengthy study material into clear, concise revision notes.
- Explain difficult concepts in simple, student-friendly language.
- Generate multiple-choice, short-answer, and descriptive practice questions.
- Build a personalised revision plan based on the topics and the student's learning needs.
- Pick out the important concepts, definitions, and key points from the supplied material.
- Give feedback and hints along the way to support active, self-paced learning.

4. Proposed GenAI Workflow

The assistant works through the student's material in a sequential pipeline, shown in Figure 1 below.

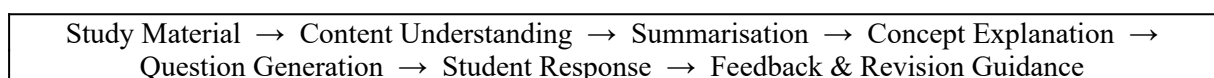


Figure 1: Proposed GenAI Workflow

Each stage builds on the one before it. The material is first read and condensed, then difficult concepts are pulled out and explained in simpler terms, and practice questions are generated

from that same content. Once the student answers, the assistant uses those responses to decide what feedback and revision guidance to give next – so the loop keeps adjusting itself to the student rather than staying static.

5. Prompt Engineering Approach

A combination of prompting techniques was used to get consistent, useful output from the model:

- Role Prompting – the model is asked to act as a patient AI tutor and explain concepts at whatever level the student requests.
- Instructional Prompting – the task, subject, difficulty level, response format, and learning objective are stated clearly, so there is little room for the model to misread the request.
- Few-Shot Prompting – a few examples of good questions, answers, summaries, or explanations are included so the model matches that style.
- Chain-of-Thought / Structured Reasoning – the model is asked to reason through the more complex educational tasks step by step internally, but only present the final, useful explanation to the student.
- Constraint-Based Prompting – limits such as word count, number of questions, difficulty level, and restriction to the supplied study material are specified explicitly.

6. Example Master Prompt

You are an expert AI tutor. Analyse the study material provided by the student and help them learn the topic effectively. First identify the key concepts and definitions. Then create a concise revision summary in simple language. Explain difficult concepts with suitable examples. Generate ten practice questions with answers, covering different difficulty levels. Do not introduce facts that are unrelated to the supplied material. If information is missing, clearly state that it is not available in the provided material.

7. Example Interactive Prompt

Act as a personal tutor. Ask me one question at a time from the selected topic. Wait for my answer before continuing. After each response, tell me whether it is correct, briefly explain the correct concept, and adjust the next question according to my performance.

8. Expected Output

Running the assistant on a given set of study material is expected to produce:

- A topic summary
- Key concepts and definitions
- Simple explanations of the harder concepts
- Important points to remember for revision
- Practice questions
- Answers with explanations
- Performance feedback
- Personalised revision recommendations

9. GenAI Technologies Used

- Large Language Models (LLMs)
- Prompt Engineering
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation (RAG), to keep responses grounded in the actual study material
- Embeddings and semantic search
- Structured output generation

10. Expected Benefits

If built out fully, this system should make studying noticeably more interactive and less time-consuming. Instead of manually preparing notes and practice material, a student can get both on demand, along with explanations and feedback whenever they are stuck. Over time, this should help students spend more of their revision time on the areas they are actually weak in, rather than re-reading material they already understand well.

11. Conclusion

Overall, this project shows how Generative AI and prompt engineering can be applied in a practical, everyday context like studying. By combining structured prompts, examples, constraints, and an interactive tutoring style, the assistant is able to turn plain study material into something personalised and genuinely useful for revision – which was the whole point of building it.